Medical Evaluation of the Oral Surgery Patient

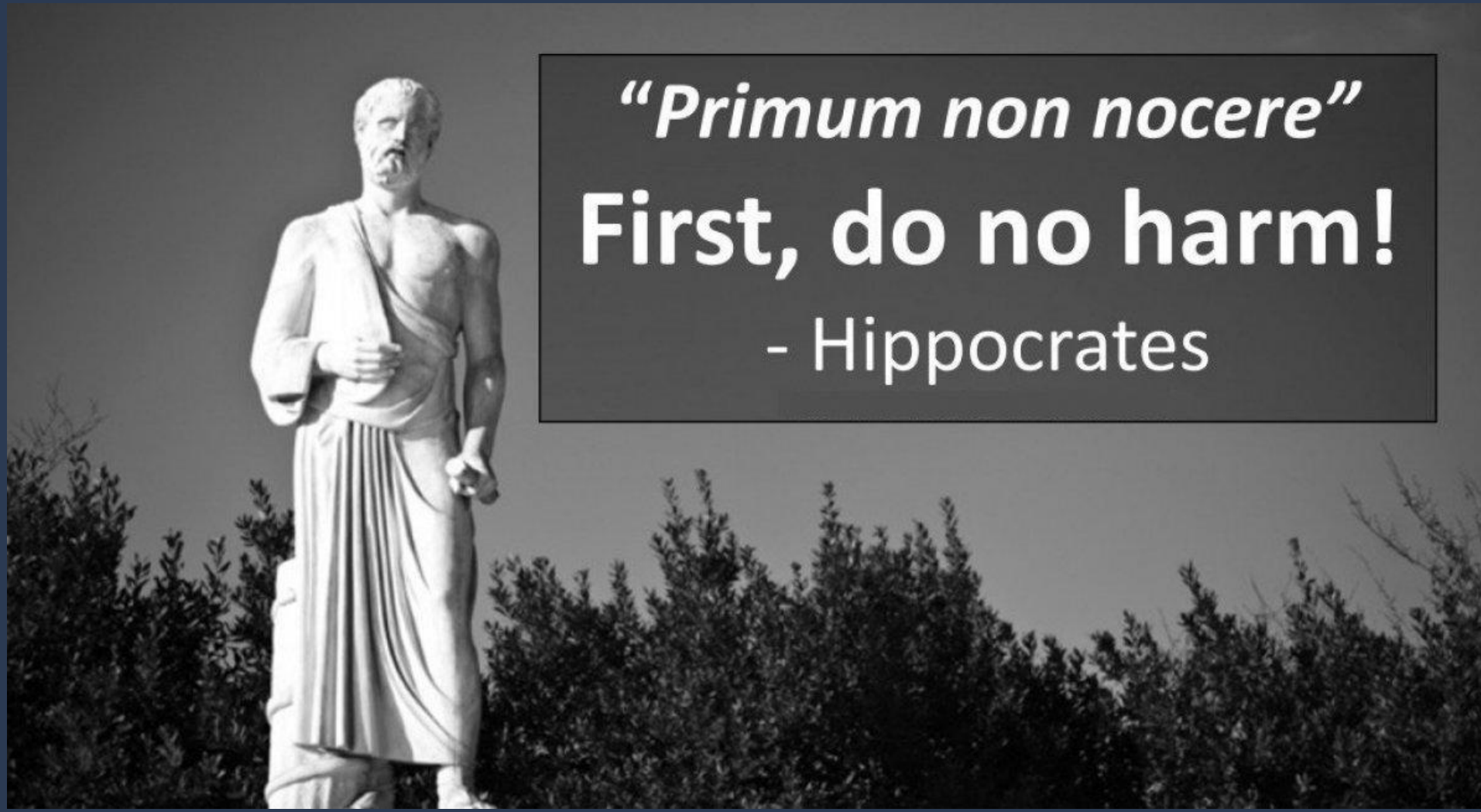
Reid Boulet
Oral and Maxillofacial Surgery

Overview

- Pre-operative patient assessment: history and physical
- Surgical considerations in common medical comorbidities
- Functional capacity and ASA classification system
- **Highlight**: Antibiotic prophylaxis guidelines
- **Highlight**: Considerations for the anticoagulated patient
- **Highlight**: Considerations in the osteoporotic patient

Goals

- Risk identification and stratification
  - Ability to tolerate procedure from a cardiorespiratory standpoint
  - Modifiable and non-modifiable risk factors
  - Identify potential peri- and post-operative pitfalls
- Medical optimization
  - Investigations
  - Pre-operative interventions to minimize known risks
- Peri-operative planning
  - Scheduling, staging
  - Anesthetic plan
  - Peri-operative medications, antibiotic prophylaxis
- Post-operative management


"Primum non nocere"
First, do no harm!
- Hippocrates

Pre-op Assessment

Who are you and what do you want?

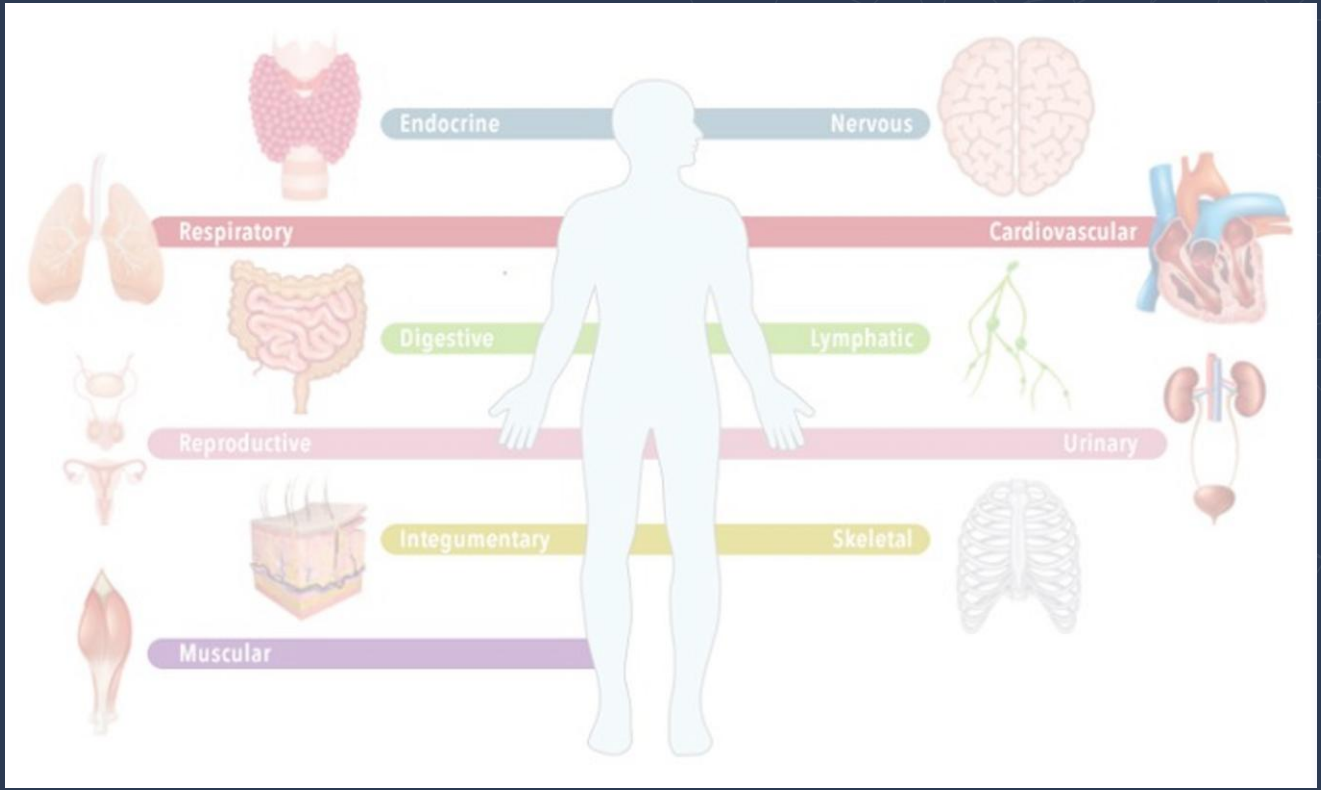
Pre-op Patient Assessment: History

- Chief complaint/History of present illness (HPI)
- Past medical history and present state of health
- Medications (past, present)
- Allergies
- Surgical history, social history, family history
- Review of systems, functional inquiry, ASA classification

Pre-op Patient Assessment: History

- *Basic demographic*
  - *"x year old man"*
- Chief complaint/History of present illness (HPI)
  - CC in their own words
  - HPI is your summary of their issue/illness
- Past medical history and present state of health
- Medications (past, present)
- Allergies
- Surgical history, social history, family history
- **Review of systems**, functional inquiry, ASA classification
- THEN PRESENT YOUR EXAM FINDINGS ("physical")
- THEN PRESENT YOUR PLAN

Review of Systems



Functional Status and ASA

- Duke Activity Status Index
  - Metabolic equivalent = rate of oxygen consumption at rest
  - 1 MET = 3.5ml/kg/min
  - VO2peak is maximum rate of oxygen consumption during exercise
  - DASI can be used to estimate VO2peak
    - VO2peak (ml/kg/min) = (0.43 x DASI) + 9.6
  - DASI score < 4 [METs] = poor functional capacity
  - Independent predictor for perioperative morbidity and mortality

Activity	METs	Functional capacity
Walk around house	1.75	Poor
Walk two blocks on level ground	2.75	Poor
Activities of daily living	2.75	Poor
Light house work (dishwashing)	2.7	Poor
Moderate housework (vacuuming)	3.5	Poor
Yard work	4.5	Moderate
Sexual relations	5.25	Moderate
Climb stairs	5.5	Moderate
Golf	6	Moderate
Swimming, basketball	7.5	Excellent
Running	8.0	Excellent

Functional Status and ASA

- ASA – Physical Classification System
  - American Society of Anesthesiologists
  - Helpful system in predicting perioperative risk
  - Multifactorial: type of surgery (length, risk, etc.), urgency of surgery, patient frailty, functional capacity and level of deconditioning

ASA Classification	Definition	Examples
ASA I	A normal healthy patient	Healthy, non-smoking, no or minimal alcohol use
ASA II	A patient with mild systemic disease	Mild diseases only without substantive functional limitations. Current smoker, social alcohol drinker, pregnancy, obesity (30<BMI<40), well-controlled DM/HTN, mild lung disease
ASA III	A patient with severe systemic disease	Substantive functional limitations; One or more moderate to severe diseases. Poorly controlled DM or HTN, COPD, morbid obesity (BMI ≥40), active hepatitis, alcohol dependence or abuse, implanted pacemaker, moderate reduction of ejection fraction, ESRD undergoing regularly scheduled dialysis, history (>3 months) of MI, CVA, TIA, or CAD/stents.
ASA IV	A patient with severe systemic disease that is a constant threat to life	Recent (<3 months) MI, CVA, TIA or CAD/stents, ongoing cardiac ischemia or severe valve dysfunction, severe reduction of ejection fraction, shock, sepsis, DIC, ARD or ESRD not undergoing regularly scheduled dialysis
ASA V	A moribund patient who is not expected to survive without the operation	Ruptured abdominal/thoracic aneurysm, massive trauma, intracranial bleed with mass effect, ischemic bowel in the face of significant cardiac pathology or multiple organ/system dysfunction
ASA VI	A declared brain-dead patient whose organs are being removed for donor purposes	

Review of Systems (and why dentists should care)

- You are treating a person, not a tooth
- You are administering medication and performing (permanent) procedures on patients!
- You CAN do harm

Cambridge child dies after dental appointment

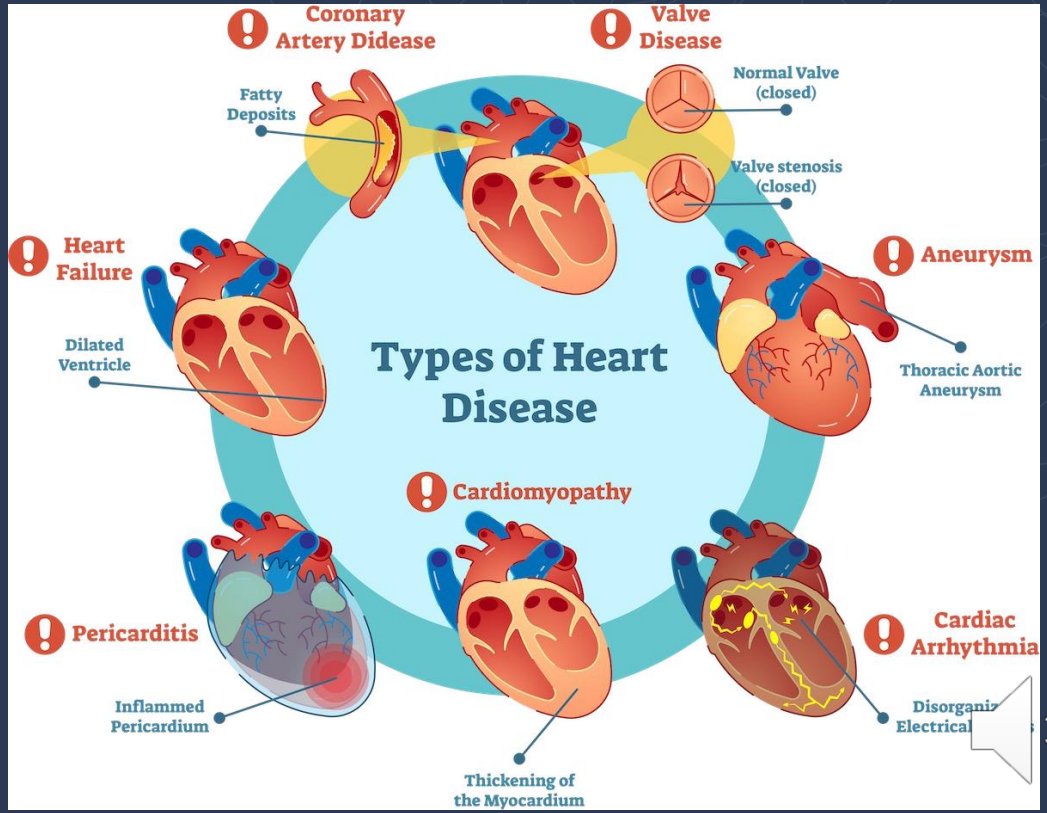
CAMBRIDGE A Cambridge child died after a visit to the dentist.

Southern Manitoba woman, 23, dies following dental procedure

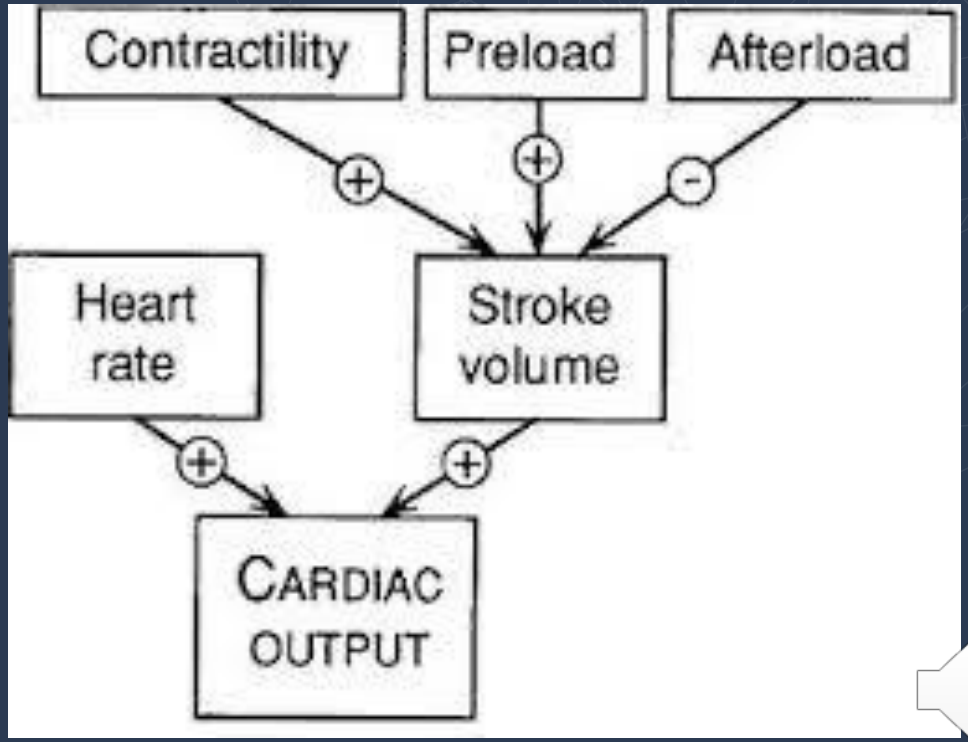
Manitoba Dental Association investigates death of woman from Morden-Winkler

Kamloops teen left with brain damage after having wisdom teeth pulled

Cardiovascular Disease



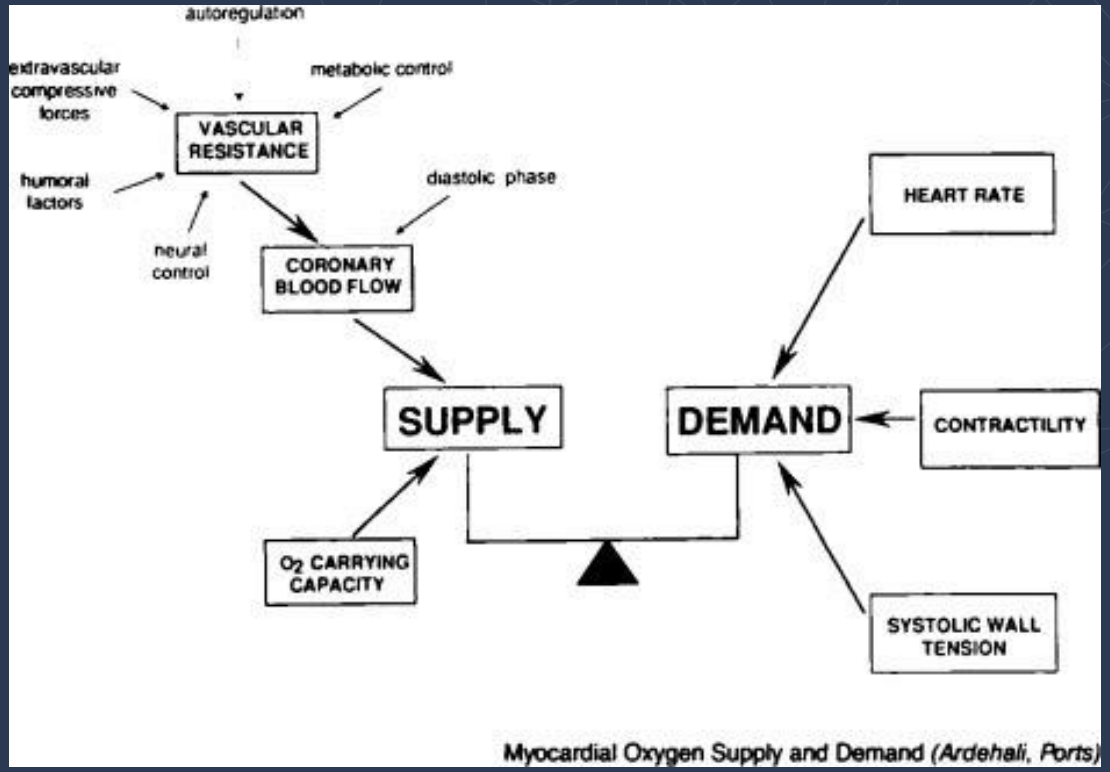
Cardiovascular Disease



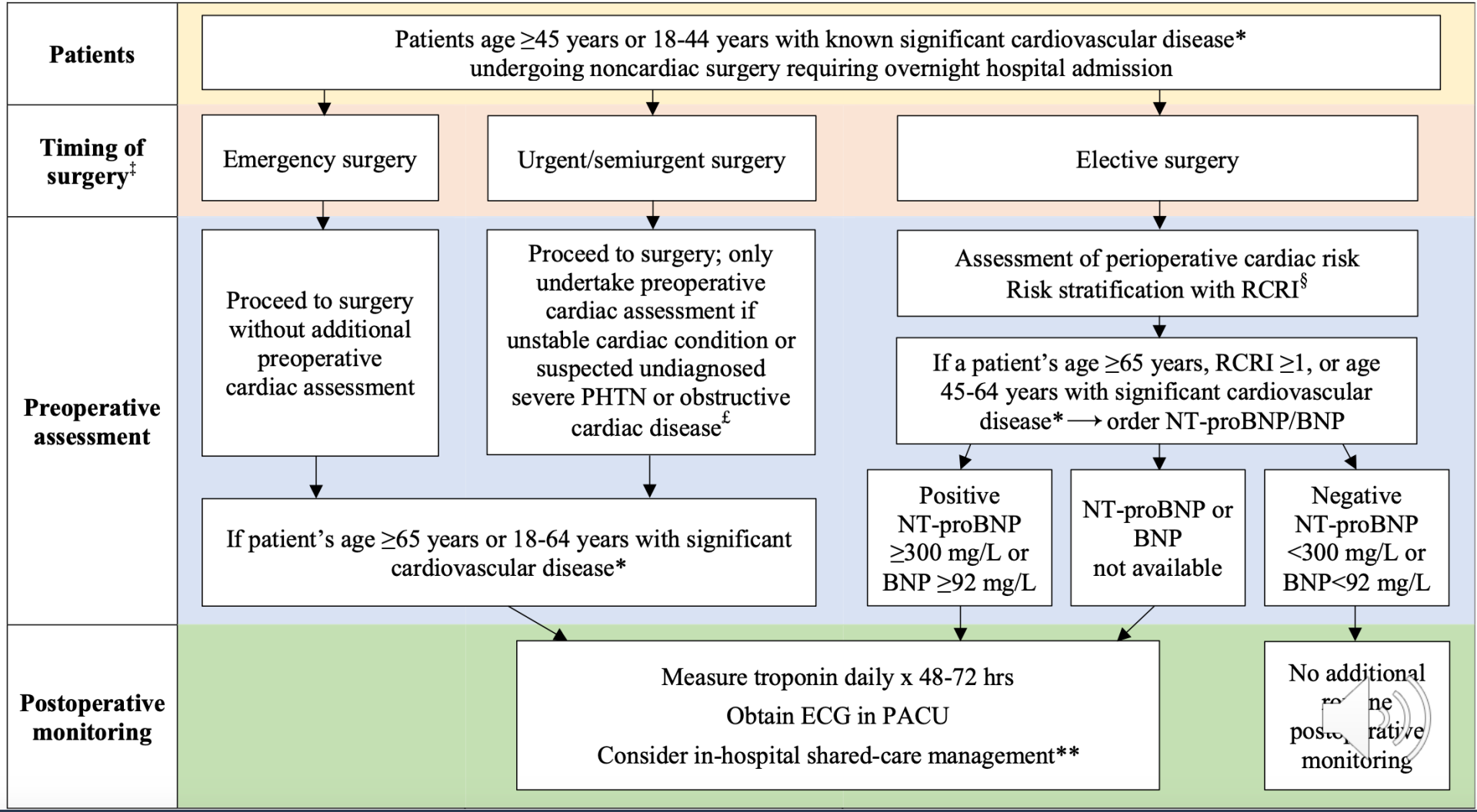
Cardiovascular Disease

- Surgical procedures induce physiologic stress (anxiety, pain epinephrine-containing anesthetics, etc.)
  - Increased sympathetic tone → increased myocardial load and oxygen demand → supply vs. demand challenge →ischemia, decompensation, etc.

Cardiovascular Disease



Myocardial Oxygen Supply and Demand *(Ardehali, Ports)*


Patients
Patients age ≥45 years or 18-44 years with known significant cardiovascular disease* undergoing noncardiac surgery requiring overnight hospital admission
Timing of surgery‡
Emergency surgery
Urgent/semiurgent surgery
Elective surgery
Preoperative assessment
Proceed to surgery without additional preoperative cardiac assessment
Proceed to surgery; only undertake preoperative cardiac assessment if unstable cardiac condition or suspected undiagnosed severe PHTN or obstructive cardiac disease£
Assessment of perioperative cardiac risk
Risk stratification with RCRI§
If a patient's age ≥65 years, RCRI ≥1, or age 45-64 years with significant cardiovascular disease* ⟶ order NT-proBNP/BNP
Positive NT-proBNP ≥300 mg/L or BNP ≥92 mg/L
NT-proBNP or BNP not available
Negative NT-proBNP <300 mg/L or BNP<92 mg/L
If patient's age ≥65 years or 18-64 years with significant cardiovascular disease*
Postoperative monitoring
Measure troponin daily x 48-72 hrs
Obtain ECG in PACU
Consider in-hospital shared-care management**
No additional routine postoperative monitoring

Cardiovascular Disease: RCRI

Table 1. Computation of Revised Cardiac Risk Index score

Variable	Points
History of ischemic heart disease*	1
History of congestive heart failure†	1
History of cerebrovascular disease‡	1
Use of insulin therapy for diabetes	1
Preoperative serum creatinine > 177 μmol/L (> 2.0 mg/dL)	1
High-risk surgery§	1

Table 2. Total RCRI score and corresponding risk of myocardial infarction, cardiac arrest, or death at 30 days after noncardiac surgery*

Total RCRI points	Risk estimate, %	95% CI for the risk estimate
0	3.9	2.8%-5.4%
1	6.0	4.9%-7.4%
2	10.1	8.1%-12.6%
≥3	15.0	11.1%-20.0%

Pulmonary Disease

- Types of pulmonary disease
  - Airway disease (ex: asthma, COPD, bronchitis, emphysema)
    - Affects the body's ability to move air in and out of the lungs
    - Obstructive → limitations in airflow due to partial or complete obstruction
    - Restrictive → reduced lung expansion and volume
  - Lung tissue disease (ex: pulmonary fibrosis, sarcoidosis)
    - Damaged tissue from injury or scarring
  - Lung circulation disease (ex: pulmonary hypertension, pulmonary edema)
    - Affects the circulation of blood to and from the heart

Pulmonary Disease

- Reduced functional capacity / exercise tolerance
- Reactive airway disease, increased secretions
- Chronic CO2 retention for obstructive disease
- Cancer risk for smokers
- Surgery/airway manipulation → exacerbations of pulmonary disease → airway compromise, respiratory distress, hypoxemia, decompensation

Pulmonary Disease Pearl

- COPD and pulmonary hypertension are among the FEW contraindications for nitrous oxide administration
  - COPD that may have alveolar bullae (air pockets or blebs) as nitrous oxide may cause bullae rupture since nitrous oxide enters that closed space more quickly than nitrogen can escape.
  - Nitrous oxide high concentrations, has a profound effect in constricting vascular smooth muscle within the lungs, thus increasing blood pressure in the pulmonary system. Patients with pre-existing pulmonary hypertension can potentially have decreases in forward blood flow and a significant cardiovascular event.

Renal Disease

- CKD=Chronic kidney disease
- 3 most common causes of CKD: HTN, DM, Glomerulonephritis
- Elevated serum creatinine and reduced creatinine clearance rate (CrCl) is indicative of abnormal kidney function
- Glomerular filtration rate is useful for staging CKD and can be estimated using CrCl (eGFR) or mathematically derived (GFR) based on a patient's serum creatinine level, age, sex and race

Renal Disease

- Reduced renal function/reduced clearance of renally excreted drugs, such as antibiotics, opioids and their metabolites, etc.
- **Consider renal dosing**
- Nephrotoxins risk further compromise of renal function
  - NSAIDs, aminoglycosides, antifungals, antivirals, antihypertensives, IV contrast media
- ESRD (end-stage renal disease) patients may be on hemodialysis – implications for appointment scheduling as they will be anticoagulated (we will cover this)

Hepatobiliary Disease

Coagulation factors synthesized by the liver
Vitamin K–dependent factors: II (prothrombin factor), VII, IX, X
Fibrinogen
Factor V
Factor VIII
Factors XI, XII, XIII
Antithrombin III
Plasminogen
Protein C and protein S

Detoxification:
- Drugs/Alcohol
- Fatty acids
- Steroid hormones
- Ammonia → Urea
- Environmental toxins/allergens

Production of Cholesterol:
- Precursor to sex hormones, Vitamin D

Storage of Micronutrients:
- Minerals: Copper, Zinc, Magnesium, Iron
- Vitamins: Vitamin A, D, E, K, B12

Metabolism:
- Conversion of T4 → T3
- Detoxification of fat

Blood Sugar Balance:
- Storage of glycogen

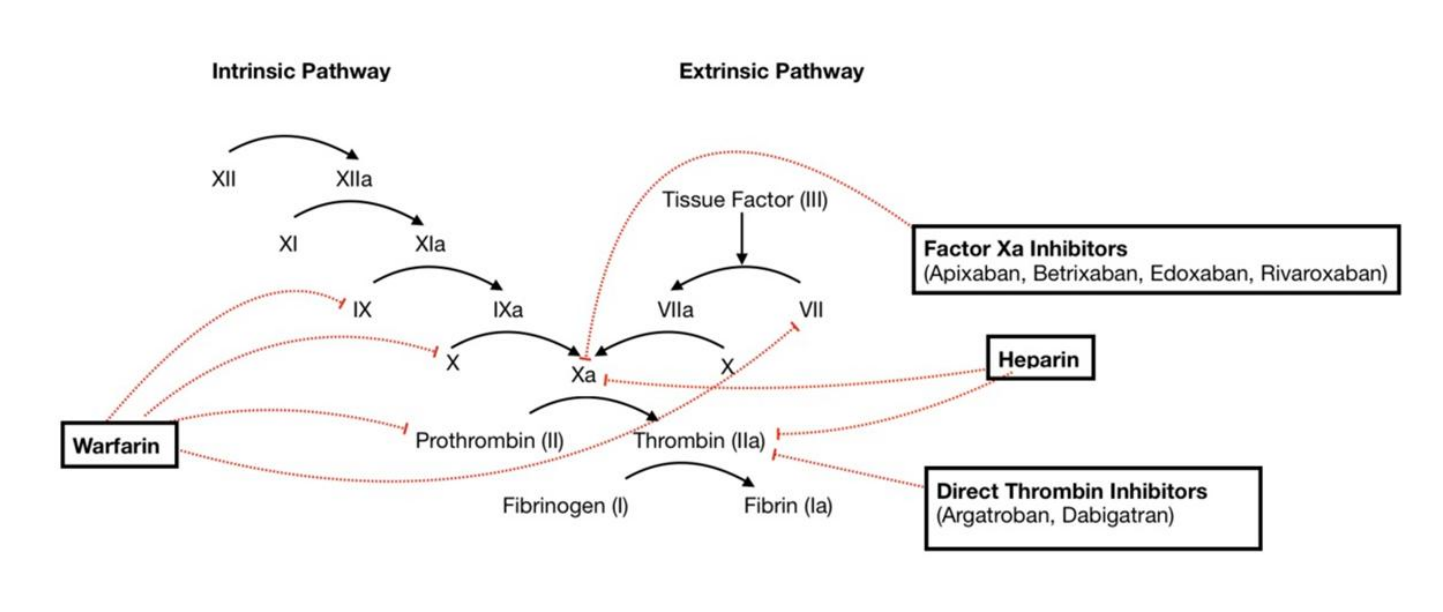
Immune System:
- Contains viruses and pathogens
- Maintenance of the hepatic and portal vein immune system

Production of Bile:
- Needed for digestion
- GI anti-microbial

Protein Synthesis:
- Blood clotting (prothrombin)
- Cholesterol transport (lipoproteins)
- Immune Function (globulins)
- Oncotic pressure (albumin)
- Copper bioavailability (ceruloplasmin)

Hepatobiliary Disease

Coagulation factors synthesized by the liver
Vitamin K–dependent factors: II (prothrombin factor), VII, IX, X
Fibrinogen
Factor V
Factor VIII
Factors XI, XII, XIII
Antithrombin III
Plasminogen
Protein C and protein S



Heptatobiliary Disease

- Liver does a lot of stuff
- Alcoholism - withdrawal, encephalopathy, seizures
- Bloodborne pathogens - HBV, HCV, HIV
- Fluid overload - hyperdynamic circulation, ascites, congestive heart failure, portal hypertension
- Electrolyte imbalances - Arrythmias
- Altered pharmacokinetics and pharmacodynamics - local anesthetics, analgesics, antibiotics, sedatives
- Coagulopathies and/or hypercoagulability

Endocrine Disease: Diabetes

- Insulin-dependent (type I) vs. non-insulin dependent/insulin resistant (type II)
- Screening for type II DM: family history, high-risk populations, history of GDM/prediabetes, CV risk factors, presence of end-organ damage associated
- with DM
- Diagnosis of diabetes
  - Glycosylated hemoglobin (HbA1c)
  - Fasting plasma glucose
  - Plasma glucose levels 2 hours after oral glucose tolerance test
- Implications of poorly controlled DM
  - Macrovascular (CVD, PVD, cerebrovascular disease)
  - Microvascular (nephropathy, retinopathy, neuropathy)
  - Immunocompromise - impaired innate and adaptive immune function

Endocrine Disease: Diabetes

Test	Prediabetes[1]	Diabetes[2]	Comments
A1C	A1C 5.7-6.4%	≥ 6.5%	A1C should be measured in a NGSP-certified laboratory. The purpose of the NGSP is to standardize HbA1c test results to those of the Diabetes Control and Complications Trial (DCCT) and United Kingdom Prospective Diabetes Study (UKPDS) which established the direct relationships between A1c levels and outcome risks in patients with diabetes.
Fasting plasma glucose (FPG)	100-125 mg/dL (5.6-6.9 mmol/L)	≥ 126 mg/dL (7.0 mmol/L)	No caloric intake for at least 8 hours
2-h plasma glucose during OGTT	140-199 mg/dL (7.8-11.1 mmol/ L)	≥ 200 mg/dL (11.1 mmol/L)	Using glucose load of 75 g anhydrous glucose dissolved in water
Random plasma glucose[3]		> 200 mg/dL (11.1 mmol/L)	Random glucose: Any time of the day without regard to time since last meal Symptoms: Polyuria, polydipsia, unexplained weight loss

Figure 1.

The 2022 American Diabetes Association Diagnostic Criteria for Prediabetes and Diabetes.

Endocrine Disease: Diabetes

- Hyper and hypo glycemic emergencies
  - DKA
  - Hypoglycemia emergency
- Exacerbation of comorbid conditions
  - Cerebrovascular accident, acute on chronic kidney failure, autonomic dysfunction
- Increased risk of postop complications
  - Delayed wound healing, surgical site infection

Endocrine Disease: Diabetes Pearls

- Just take the blood sugar

Hematologic Disease

- Anemia
- Coagulopathies → bleeding!
- Hypercoagulable states (not enough bleeding!)
  - Venous thromboembolism (VTE) ie. DVT, PE
  - Systemic thromboembolism ie. ischemic stroke, myocardial infarct, acute limb ischemia
- Hematologic malignancy
  - Many considerations re: hematologic function, immune function
  - Multiple myeloma – must ask about IV bisphosphonate history

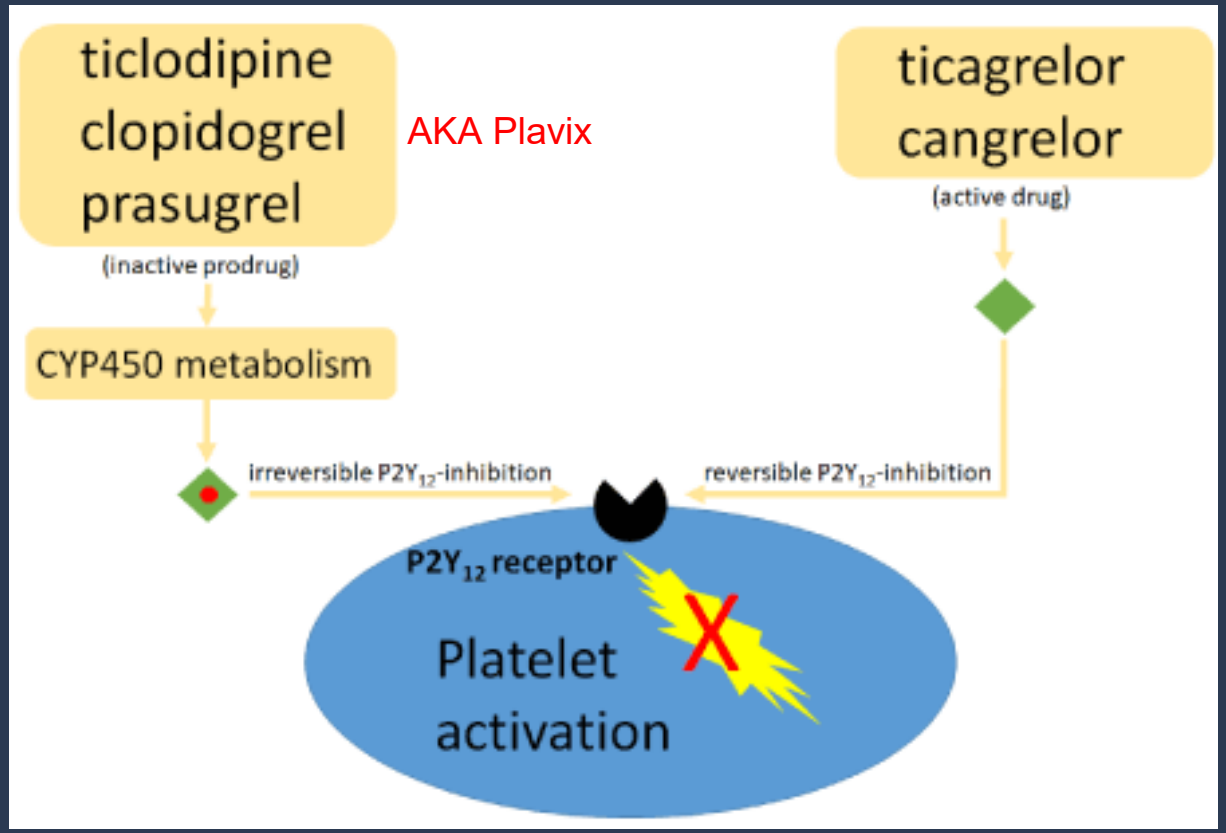
Hematologic Disease Pearls

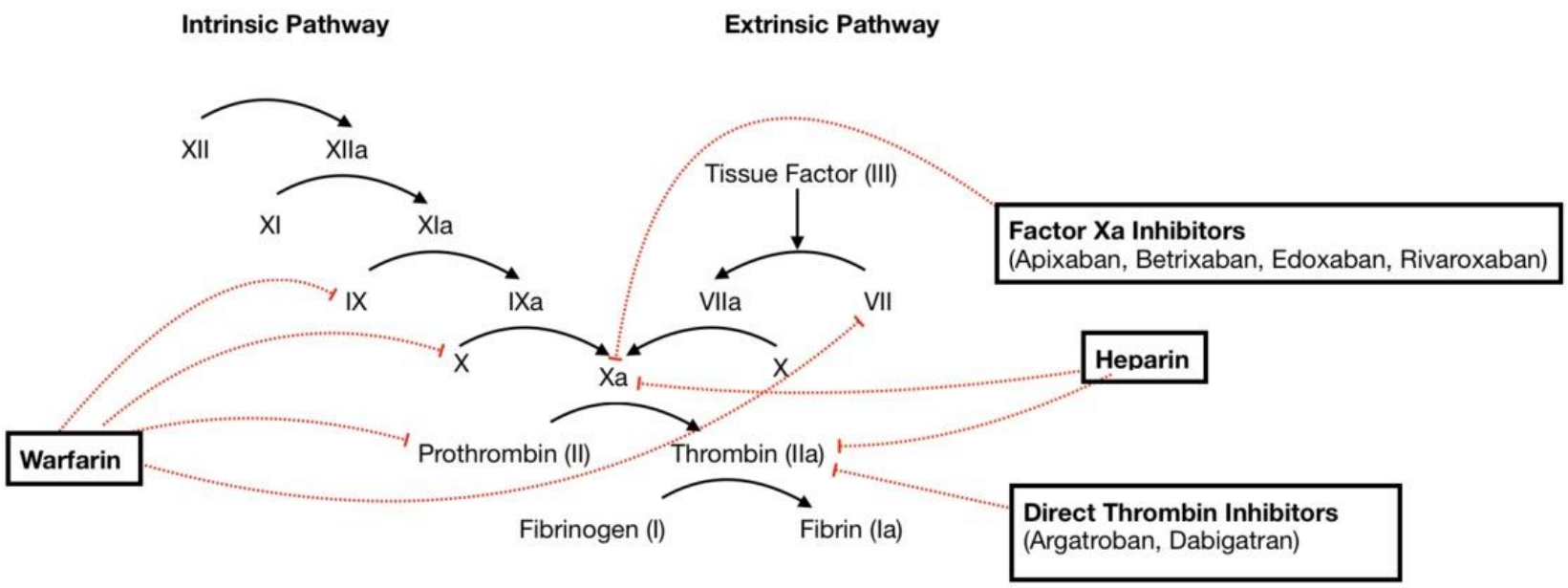
- Not all “blood thinners” are made equal..

As aside on the Anticoagulated patient

What medications again?

- Anti **coagulants** (interrupt coagulation cascade)
  - Used in low-velocity flow states (Deep Vein Thrombosis, Pulmonary Embolism, stroke prophylaxis in Atrial Fibrillation)
  - Used for mechanical heart valves: Ex. Warfarin (Vitamin K antagonist)
- Anti **platelets** (COX inhibitor, P2Y12 inhibitor)
  - Used in high-velocity flow states (Myocardial Infarction, Cerebrovascular Accident, Peripheral Arterial Disease, etc)
  - Dual antiplatelet therapy=DAPT
    - Indications are serious:
    - Post-ACS event
    - Stable Ischemic heart disease (SIHD)
    - Severe diffuse atherosclerosis


ticlodipine
clopidogrel
prasugrel
AKA Plavix
ticagrelor
cangrelor
(active drug)
(inactive prodrug)
CYP450 metabolism
irreversible P2Y12-inhibition
reversible P2Y12-inhibition
P2Y12 receptor
Platelet
activation



DOAC=Direct oral anticoagulants (DOACs)
Factor Xa inhibitors are in the name!
Apixaban, rivaroxaban

Why not start anticoagulants?

- 8-10% case fatality of major bleeding events
- HAS-BLED score: estimates risk of major bleeding for patients on WARFARIN

Condition	Points
H - Hypertension	1
A - Abnormal renal or liver function (1 point each)	1 or 2
S - Stroke	1
B - Bleeding	1
L - Labile INRs	1
E - Elderly (> 65 years)	1
D - Drugs or alcohol (1 point each)	1 or 2

HAS-BLED score	Bleeds per 100 patient-years
0	1.13
1	1.02
2	1.88
3	3.74
4	8.70
5	12.5

Note: HAS-BLED has been validated for warfarin, but not for the new anticoagulants.

Why not stop anticoagulants?

- Risk of thromboembolic events
  - Mechanical heart valve thrombosis: 15% fatality rate
  - Embolic stroke: death or major disability in 70% of patients
  - Venous thromboembolism (i.e. DVT, PE)
    - 5-9% fatality
- Who is at risk? Virchow's triad"
- Risk stratification schema for predicting TE events in uncoagulated patients with non-valvular atrial fibrillation include CHADS2, CHA2DS2-VASc

TABLE 2. PATIENT STRATIFICATION FOR THROMBOEMBOLISM RISK

HIGH THROMBOEMBOLIC RISK *(BRIDGING ANTICOAGULATION SUGGESTED)*:

- Any mechanical prosthetic mitral valve
- Older generation (cage-ball, tilting disc) mechanical aortic valve
- Chronic atrial fibrillation (valvular or non-valvular) with a $CHADS_2$ score* of 5-6
- Recent (within 3 months) arterial thromboembolism (stroke, systemic embolism, transient ischemic attack [TIA])
- Recent (within 3 months) venous thromboembolism (deep vein thrombosis, pulmonary embolism)†
- Prior arterial or venous thromboembolism during appropriate interruption of warfarin
- Severe thrombophilia with history of venous thromboembolism (e.g. deficiency of protein C, protein S or antithrombin, antiphospholipid syndrome)
- Rheumatic valvular heart disease

INTERMEDIATE THROMBOEMBOLIC RISK *(BRIDGING ANTICOAGULATION OPTIONAL AND BASED ON INDIVIDUAL PATIENT CHARACTERISTICS)*:

- Chronic atrial fibrillation with a $CHADS_2$ score* of 3-4
- Newer generation (bileaflet) mechanical aortic valve
- Prior arterial or venous thromboembolism within last 3-12 months

LOW-RISK *(BRIDGING ANTICOAGULATION IS NOT RECOMMENDED)*:

- Chronic atrial fibrillation (valvular or non-valvular) with a $CHADS_2$ score* of 0-2
- Prior venous thromboembolism over 12 months ago
- Bioprosthetic heart valve

*$CHADS_2$ score estimates the risk of stroke in patients with non-valvular atrial fibrillation. The score is the total points for the presence of congestive heart failure (1), hypertension (1), age $\geq$ 75 yrs (1), diabetes (1), stroke, transient ischemic attack or systemic embolism (2).

	Score	$CHADS_2$ score	Adjusted stroke rate (%/year)
$CHADS_2$ acronym			
Congestive heart failure	1	0	1.9%
Hypertension	1	1	2.8%
Aged $\geq$ 75 years	1	2	4.0%
Diabetes mellitus	1	3	5.9%
Stroke/TIA	2	4	8.5%
Max score	6	5	12.5%
		6	18.2%
CHA_2DS_2-VASc acronym			
Congestive heart failure/ LV dysfunction	1	0	0%
Hypertension	1	1	0.7%
Aged $\geq$ 75 years	2	2	1.9%
Diabetes mellitus	1	3	4.7%
Stroke/TIA/TE	2	4	2.3%
Vascular disease (prior to MI, PAD, or aortic plaque)	1	5	3.9%
Aged 65–74 years	1	6	4.5%
Sex category (ie, female gender)	1	7	10.1%
Max score	10	8	14.2%
		9	100%

Abbreviations: LV, left ventricular; MI, myocardial infarction; PAD, peripheral artery disease; TE, thromboembolism; TIA, transient ischemic attack.

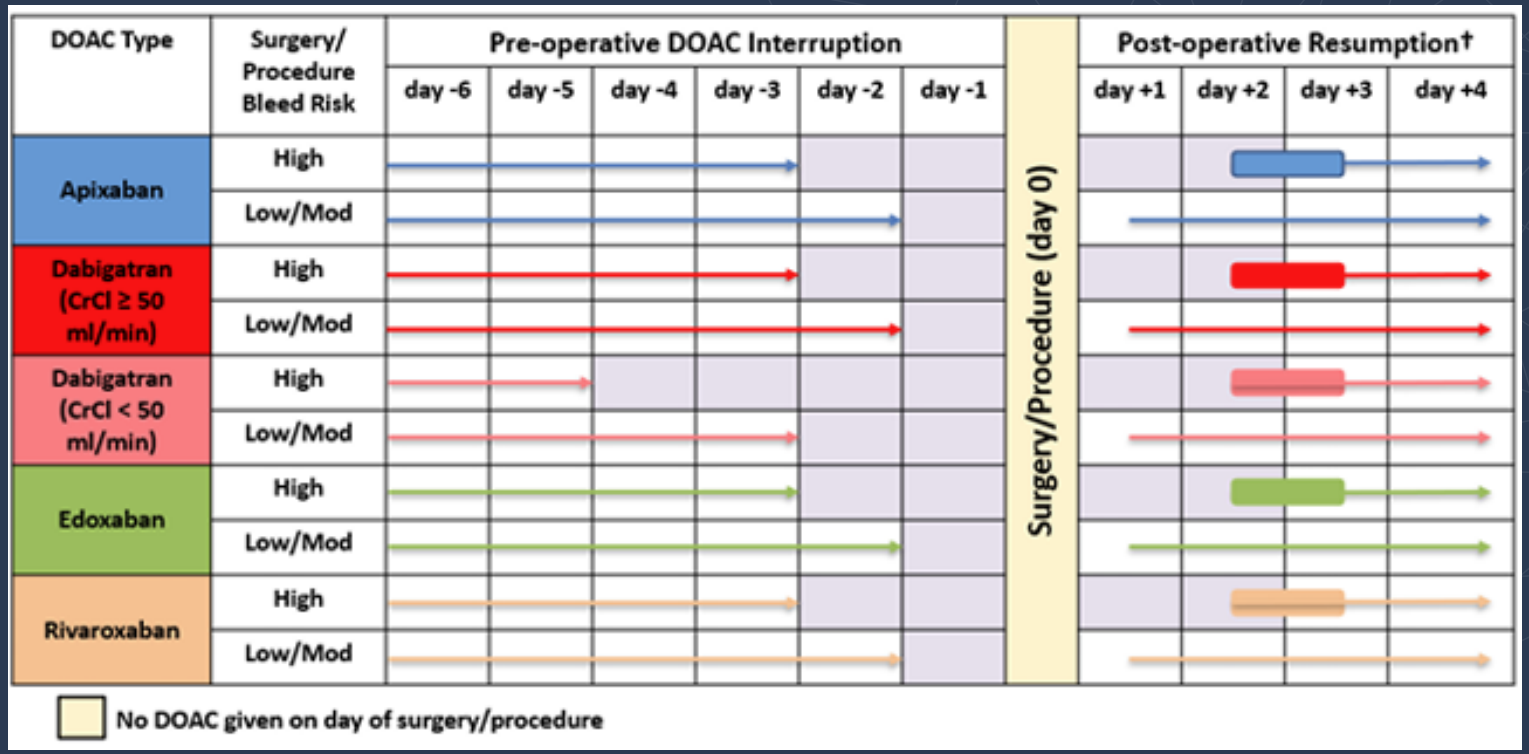
Peri-Operative Management of Patients who are Receiving Warfarin

Thrombosis Canada
Thrombose Canada

LOW/VERY LOW RISK	MODERATE RISK	HIGH RISK
• Dental extractions (1 or 2 teeth), endodontic (root canal) procedure, • Subgingival scaling or other cleaning • Cataract surgery • Dermatologic procedures (e.g. biopsy) • Gastroscopy or colonoscopy without biopsies • Coronary angiography • Permanent pacemaker insertion or internal defibrillator placement (if bridging anticoagulation is not used) • Selected procedures (e.g. thoracentesis, paracentesis, arthrocentesis)	• Other intra-abdominal surgery (e.g. laparoscopic cholecystectomy, hernia repair, colon resection) • Other general surgery (e.g. breast) • Other intrathoracic surgery • Other orthopedic surgery • Other vascular surgery • Non-cataract ophthalmologic surgery • Gastroscopy or colonoscopy with biopsies • Selected procedures (e.g. bone marrow biopsy, lymph node biopsy) • Complex dental procedure (e.g. multiple tooth extractions)	• Any surgery or procedure with neuraxial (spinal or epidural) anesthesia • Neurosurgery (intracranial or spinal) • Cardiac surgery (e.g. CABG, heart valve replacement) • Major intra-abdominal surgery (e.g. intestinal anastomosis) • Major vascular surgery (e.g. aortic aneurysm repair, aortofemoral bypass) • Major orthopedic surgery (e.g. hip or knee replacement) • Lung resection surgery • Urological surgery (e.g. prostatectomy, bladder tumour resection) • Extensive cancer surgery (e.g. pancreas, liver) • Reconstructive plastic surgery • Selected procedures (e.g. kidney biopsy, prostate biopsy, cervical cone biopsy, pericardiocentesis, colonic polypectomy)

Warfarin Recommendations

- Extractions: Obtain INR within 24 hours of procedure
- INR < 4.0 can undergo extractions without interruption
  - Recall: INR therapeutic target generally 2.0 - 3.0
- Treat extraction sites with local measures (i.e. gel foam, surgical, sutures)
- INR > 4.0 likely supra-therapeutic
- Refer to MRP for optimization
- More advanced procedures (i.e. Maxillofacial trauma or reconstruction, resections…)
  - Heparin bridging vs. no-bridging protocols



For moderate risk procedures

Antiplatelet Recommendations

jcda.ca

Dental Management of Patients Undergoing Antithrombotic Therapy

Published by the Canadian Dental Association

J Can Dent Assoc. 2020;86:k17

Antiplatelet agent Recommendations

"Current recommendations do not suggest discontinuing the use of daily aspirin before routine dental extractions, including multiple routine extractions.[31] Recommendations also state that the thromboembolic risk of discontinuing aspirin and clopidogrel likely outweighs potential bleeding complications associated with surgery.[8]

PERIOPERATIVE MANAGEMENT OF ANTIPLATELET THERAPY

Thrombosis Canada
Thrombose Canada

DIAGNOSTIC TESTING, ARTHROCENTESIS, AND MINOR DENTAL, SKIN AND EYE PROCEDURES

Patients undergoing arthrocentesis, minor dental (extraction, root canal), eye (cataract) or skin (biopsy, skin cancer excision) procedures, as well as low bleeding risk diagnostic procedures, can continue ASA without interruption. Less is known about the safety of continuing P2Y12 inhibitors (clopidogrel, ticagrelor, prasugrel) around minor procedures when taken as monotherapy. It is reasonable to discontinue them for a short period (3-4 days) before the procedure. If patients are also taking ASA (dual antiplatelet therapy), the P2Y12 should be discontinued prior to surgery (clopidogrel and ticagrelor for 5-7 days, prasugrel for 7-10 days) while ASA is continued.

Antiplatelet Recommendations

	Timing of Non-cardiac Surgery	Perioperative Antiplatelet Management
PCI Patients with a Bare Metal Stent	Recommended to delay surgery for at least 1 month after PCI	ASA should be continued perioperatively. Clopidogrel and ticagrelor should be withheld 5-7 days preoperatively, and prasugrel 7-10 days preoperatively
		P2Y12 inhibitor should be restarted as soon as it is deemed safe by the surgeon
PCI Patients with a Drug Eluting Stent	Recommended to delay surgery for at least 3 months after PCI. If semi-urgent surgery is required, surgery should be delayed at least 1 month after PCI with a DES.	ASA should be continued perioperatively. Clopidogrel and ticagrelor should be withheld 5-7 days preoperatively, and prasugrel 7-10 days preoperatively
		P2Y12 inhibitor should be restarted as soon as it is deemed safe by the surgeon

"Moderate bleed risk procedures"

Antiplatelet Recommendations

- Minimal bleed risk procedures
  - Continue in perioperative period
  - Defer Daily dose 4-6hrs post op
- Moderate bleed risk procedures (i.e. more extensive dentoalveolar, ex: FMC)
  - Discontinue DOAC for 2-3 half-lives pre-op (i.e. hold dabigatran 2d preop if normal kidney function)
  - Discontinue DOAC for 4-5 half-lives preop (i.e. hold dabigatran 3d preop if normal kidney function
- Bottom line: never discontinue aspirin
- Always remember the risk vs. reward
- Consider consultation patient physician

As aside on the Osteoporotic patient

MRONJ

The feared complication

Diagnostic Criteria

- History of exposure to bisphosphonates or denosumab or antiangiogenic medications
- No history of radiation
- Exposed bone of the jaw for > 8 weeks

Journal of
ORAL AND
MAXILLOFACIAL SURGERY

American Association of Oral and Maxillofacial Surgeons' Position Paper on Medication-Related Osteonecrosis of the Jaws—2022 Update

Salvatore L. Ruggiero, DMD, MD, [*] *Thomas B. Dodson, DMD, MPH,* [†] *Tara Aghaloo, DDS, MD, PhD,* [‡] *Eric R. Carlson, DMD, MD, EdM,* [§] *Brent B. Ward, DDS, MD,* [‖] *and Deepak Kademani, DMD, MD* [¶]

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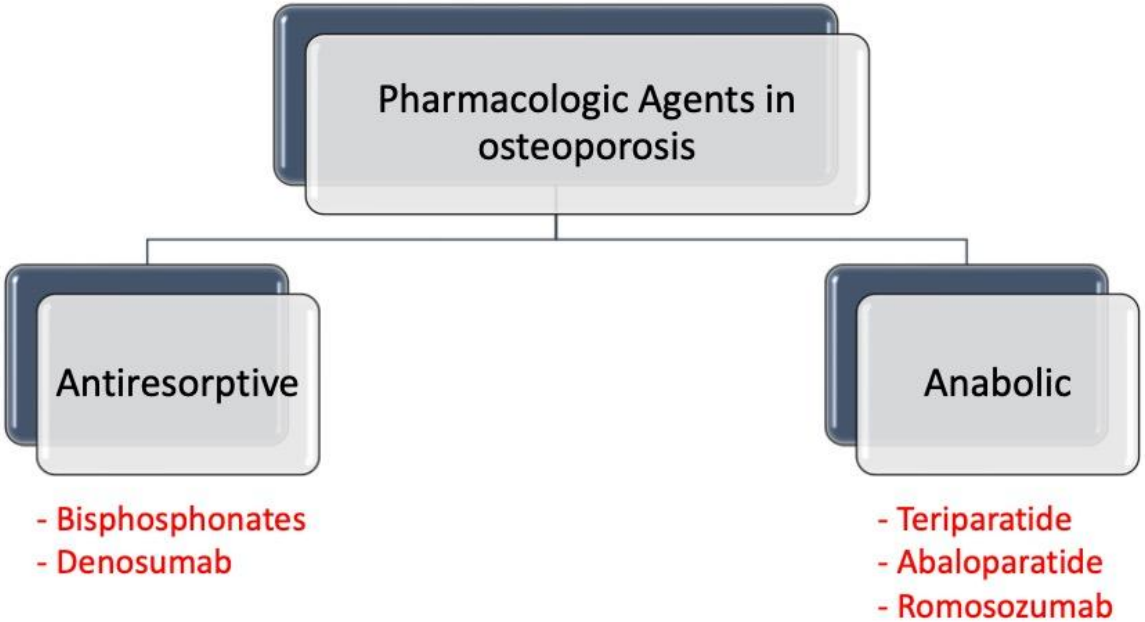
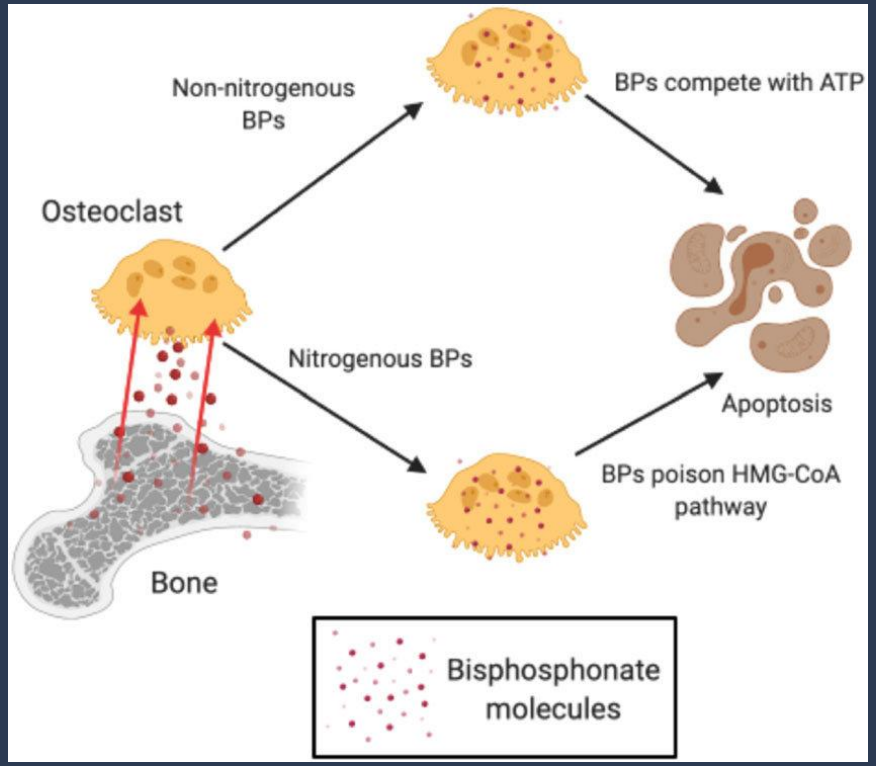
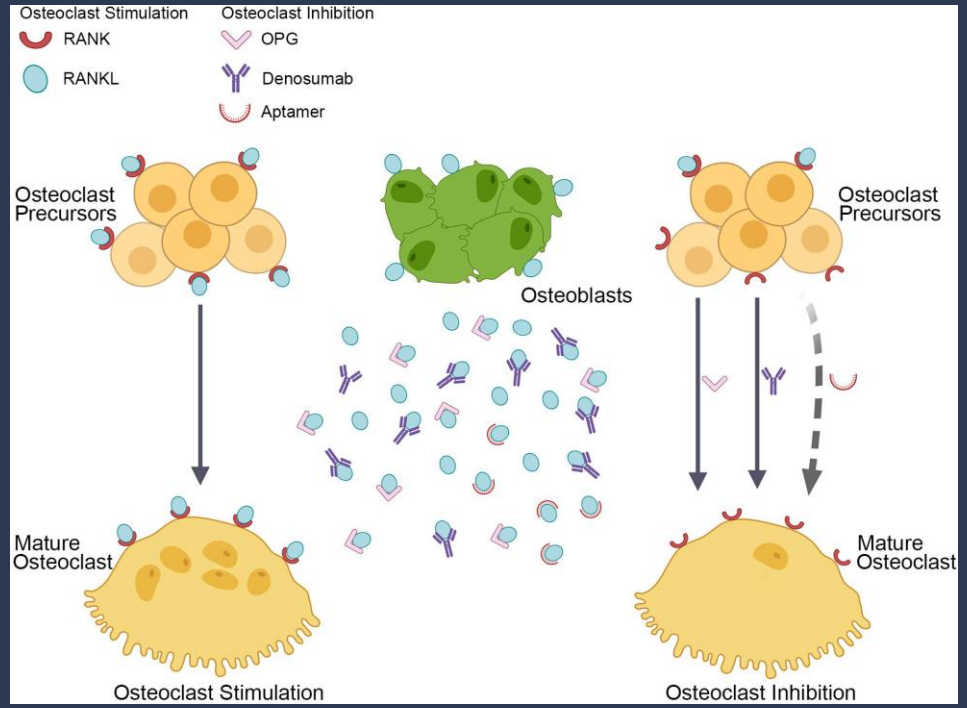


Antiresorptives

- Indications:
  - Prevention of skeletal related events (SRE) in metastatic bone disease or multiple myeloma
    - Pathologic fractures, spinal cord compression, hypercalcemia, bone pain
  - Prevention of fragility fractures in metabolic bone disease (i.e., osteoporosis, Paget's disease, osteogenesis imperfecta

Bisphosphonates vs. Rank Ligand Inhibitors (Denosumab)





Antiresorptives: who is at risk for MRONJ?

- Medication-related factors
  - Indication for antiresorptive therapy (OP vs Malignancy)
  - Agent (oral vs IV bisphosphonate vs denosumab)
  - Duration of antiresorptive therapy
- Local factors
  - Dentoalveolar surgery (especially extractions or implants)
  - Anatomic factors (75% md, 25% max, 4.5% both jaws)
  - Concomitant oral disease (periodontal or periapical pathosis)
- Demographic and Systemic factors
  - Older age, female gender
  - Immunosuppressants (corticosteroids, DMARDs, numerous chemotherapeutics)
  - Anemia
  - Cancer
- • Tobacco use (controversial correlation to MRONJ)

Antiresorptives and Extractions

- Risk of post extraction MRONJ in osteoporotic patient
  - Oral bisphosphonates: 0-0.15%
  - Denosumab: ~ 1%
- Risk of post extraction MRONJ in cancer patient
  - **IV bisphosphonates: 1.6-14.8%**
  - **Denosumab: similar?**
- Drug holiday?
  - Multifactorial decision (indications/type antiresorptives, comorbidities, medications, underlying infection/inflammation, extent of surgery required)
  - Bisphosphonates: 2022 AAOMS update unable to reach a consensus
  - Denosumab: drug holiday is rational and justifiable based on pharmacokinetics
    - Defer extractions 3-4 months since last dose -> procedure -> 6-8 weeks healing > Resume

Zolendronate (Zometa®) (Reclast®)	Bone Metastases Osteoporosis	Yes	4 mg/3 weeks 5 mg/year	IV IV
Denosumab (Xgeva®) (Prolia®)	Bone metastases Osteoporosis	No Humanized monoclonal antibody	120 mg/4 weeks 60 mg/6 months	SQ SQ

Antiresorptives and Extractions

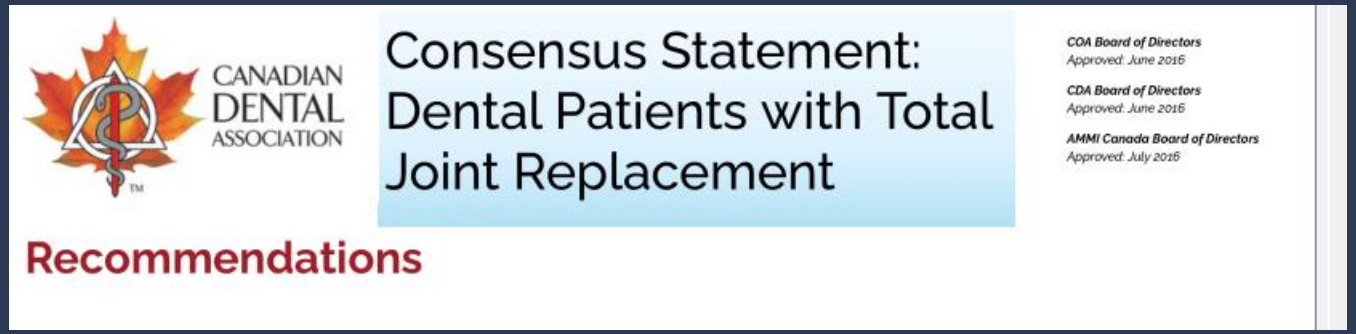
- Prevention includes antibiotic prophylaxis, “good oral hygiene”
- No specific guidelines

An aside on Antibiotic Prophylaxis

Get me amoxicillin, STAT

Joints

- No evidence that antibiotic prophylaxis has any benefit in joint replacement patients



Infective Endocarditis Prophylaxis: 2021 revision

- Summary:
  - Available evidence suggests these same 4 conditions are at HIGH RISK for acquiring IE and also HIGHEST RISK for adverse outcomes secondary to IE
  - Antibiotic prophylaxis continues to be recommended in these high-risk groups
  - Moderate lifetime risk conditions such as rheumatic heart disease, congenital valve abnormalities have similar mortality rates due to IE but data is NOT organism specific (i.e. VGS) à cannot conclude that risk of antibiotic prophylaxis > potential benefit
  - Expansion on conditions requiring antibiotic prophylaxis is not supported

Infective Endocarditis Prophylaxis: 2021 revision

Antibiotic Prophylaxis Suggested
Prosthetic cardiac valve or material
Presence of cardiac prosthetic valve
Transcatheter implantation of prosthetic valves
Cardiac valve repair with devices, including annuloplasty, rings, or clips
Left ventricular assist devices or implantable heart
Previous, relapse, or recurrent IE
CHD
Unrepaired cyanotic congenital CHD, including palliative shunts and conduits.
Completely repaired congenital heart defect with prosthetic material or device, whether placed by surgery or by transcatheter during the first 6 mo after the procedure
Repaired CHD with residual defects at the site of or adjacent to the site of a prosthetic patch or prosthetic device
Surgical or transcatheter pulmonary artery valve or conduit placement such as Melody valve and Contegra conduit
Cardiac transplant recipients who develop cardiac valvulopathy

Antibiotic Prophylaxis Not Suggested
AP for a dental procedure not suggested
Implantable electronic devices such as a pacemaker or similar devices
Septal defect closure devices when complete closure is achieved
Peripheral vascular grafts and patches, including those used for hemodialysis
Coronary artery stents or other vascular stents
CNS ventriculoatrial shunts
Vena cava filters
Pledgets

AP suggested
All dental procedures that involve manipulation of gingival tissue or the periapical region of teeth or perforation of the oral mucosa
AP not suggested
Anesthetic injections through noninfected tissue, taking dental radiographs, placement of removable prosthodontic or orthodontic appliances, adjustment of orthodontic appliances, placement of orthodontic brackets, shedding of primary teeth, and bleeding from trauma to the lips or oral mucosa

Infective Endocarditis Prophylaxis: 2021 revision

2021 Antibiotic Prophylaxis Recommendations: Single dose 30-60min before procedure

Situation	Agent	Adults	Children
Oral	Amoxicillin	2 g	50 mg/kg
Unable to take oral medication	Ampicillin OR	2 g IM or IV	50 mg/kg IM or IV
	Cefazolin or ceftriaxone	1 g IM or IV	50 mg/kg IM or IV
Allergic to penicillin or ampicillin—oral	Cephalexin*† OR	2 g	50 mg/kg
	Azithromycin or clarithromycin OR	500 mg	15 mg/kg
	Doxycycline	100 mg	<45 kg, 2.2 mg/kg >45 kg, 100 mg
Allergic to penicillin or ampicillin and unable to take oral medication	Cefazolin or ceftriaxone†	1 g IM or IV	50 mg/kg IM or IV

Pearls:

- If AP is inadvertently missed before procedure, it may be administered up to 2hrs post procedure
- Most people are not allergic to penicillin

Soap Box

Take care of your patients
Ask questions
Don’t kill anyone please

Thank You

Questions?

Reid.boulet@mail.utoronto.ca

Supplemental

Anemia ie:

Microcytic

- Iron defic
- sideroblastic, Thalassemias
- ACD

Normocytic

- Aplastic anemia,
- Hemolytic (ie sickle)
- Hemorrhage

Macrocytic

- B12 deficiency
- Folate deficiency
- Fanconi anemia

Coagulopathies ie:

Iatrogenic

- **Antiplatelets**
- **Vitamin K antags**
- **DOACs**
- **Heparin**

Hereditary

- Hemophilia A
- Hemophilia B
- von Willebrands disease

Acquired

- Liver disease
- HUS, ITP, TTP, DIC, HELLP, HITT, VITT, Bone marrow Failure, etc..

Hypercoagulable states ie:

- Protein C or S deficiency
- Antithrombin III deficiency
- Antiphospholipid antibody syndrome
- Sickle cell disease

Hematologic malignancies and related disorders…

- Myeloid malignancies
- Myeloproliferative dz
- Lymphoid malignancies
- Malignant Clonal Proliferations

Stages of CKD of all types		
Stage	**Qualitative Description**	**GFR (mL/min/1.73 m²)**
1	Kidney damage – normal GFR	> 90*
2	Kidney damage – mild ↓ GFR	60-89*
3a	Moderate ↓ GFR	45-59
3b	Moderate ↓ GFR	30-44
4	Severe ↓ GFR	15-29
5	End-stage renal disease	<15

*A GFR >60 mL/min/1.73 m^2 in isolation is not CKD, unless other evidence of kidney damage is present

CKD, chronic kidney disease; *GFR*, glomerular filtration rate